



Reconstruction of past climate conditions over central Europe from groundwater data

J.A. Corcho Alvarado^{a,*}, M. Leuenberger^b, R. Kipfer^{c,d}, T. Paces^e, R. Purtschert^b

^a Institute of Radiation Physics, University of Lausanne and CHUV, Rue de Grand-Pré 1, 1007 Lausanne, Switzerland

^b Climate and Environmental Physics, Physics Institute and Oeschger Centre for Climate Change Research, University of Bern, Sidlerstrasse 5, 3012 Bern, Switzerland

^c EAWAG, Department of Water Resources and Drinking Water, Swiss Federal Institute of Aquatic Science and Technology, Überlandstrasse 133, 8600 Dübendorf, Switzerland

^d ETH Zurich, Institute of Geochemistry and Petrology, Swiss Federal Institute of Technology, Mineral Resources, 8092 Zurich, Switzerland

^e Czech Geological Survey, Klarov 3, 118 21 Praha 1, Czech Republic

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ABSTRACT

Here we present a 30 000 years low-resolution climate record reconstructed from groundwater data. The investigated site is located in the Bohemian Cretaceous Basin, in the corridor between the Scandinavian ice sheet and the Alpine ice field. Noble gas temperatures (NGT), obtained from groundwater data, preserved multicentennial temperature variability and indicated a cooling of at least 5–7 °C during the last glacial maximum (LGM). This is further confirmed by the depleted $\delta^{18}\text{O}$ and $\delta^2\text{H}$ values at the LGM. High excess air (ΔNe) at the end of the Pleistocene is possibly related to abrupt changes in recharge dynamics due to progression and retreat of ice covers and permafrost. These results agree with the fact that during the LGM permafrost and small glaciers developed in the inner valleys of the Giant Mountains (located in the watershed of the aquifers). A temporal decrease of deuterium excess from the pre-industrial Holocene to present days is linked to an increase of the air temperatures, and probably also to an increase of water pressure at the source region of precipitation over the past few hundred years.

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1. Introduction

Groundwater recharge temperatures obtained from dissolved atmospheric noble gases (Noble gas temperatures: NGT) is one source of information that could significantly contribute to our understanding of large-scale, multicentennial air temperature trends. The estimation of the NGT is based on the well-known temperature-dependent solubility of the noble gases in water, and it reflects the average soil temperature at the recharge depth of groundwater (Stute and Schlosser, 1993; Aeschbach-Hertig et al., 1999, 2000; Klump et al., 2007). Variations in NGTs reflect then soil temperature changes, which generally are coupled to surface air temperature changes (Stute and Schlosser, 1993; Holocher et al., 2002; Beyerle et al., 2003; Klump et al., 2007; Althaus et al., 2009; Corcho Alvarado et al., 2009; Cey et al., 2009).

Contrary to other climate reconstruction procedures, as noble gas solution is a simple physical process and therefore calibration

against the instrumental record is not needed, it provides an independent physically-based measurement of past temperatures (Kipfer et al., 2002). This approach could offer a singular view of the amplitude of multicentennial climate variations during the Holocene, as it has been shown in recent studies (Castro et al., 2007; Corcho Alvarado et al., 2009), but more important this method captures major temperature shifts since the last glacial maximum (LGM) (Rudolph et al., 1984; Stute and Deák, 1989; Beyerle et al., 1998; Zuber et al., 2000, 2004; Kipfer et al., 2002; Blaser et al., 2010).

The interpretation of the noble gases dissolved in groundwater enables as well the quantification of the so-called “excess air” in groundwater (Kipfer et al., 2002). The “excess air” is formed by the air entrapment within the quasi-saturated zone under increased hydrostatic pressure (Holocher et al., 2002), and its magnitude depends on the recharge conditions i.e. intensity of precipitation, climatic conditions, infiltration mechanism, etc. (Klump et al., 2007). Variation of this parameter has been interpreted in terms of changes of the recharge dynamics, and therefore, of humidity conditions (Klump et al., 2008a). In regions that were ice covered, increased portions of excess air has been interpreted as an input of meltwater which is typically enriched in gases (Wilson and McNeill, 1997; Vaikmaa et al., 2001).

* Corresponding author. Tel.: +41 213148173; fax: +41 213148299.

E-mail addresses: jose-antonio.corcho-alvarado@chuv.ch (J.A. Corcho Alvarado), leuenberger@climate.unibe.ch (M. Leuenberger), Purtschert@climate.unibe.ch (R. Purtschert).

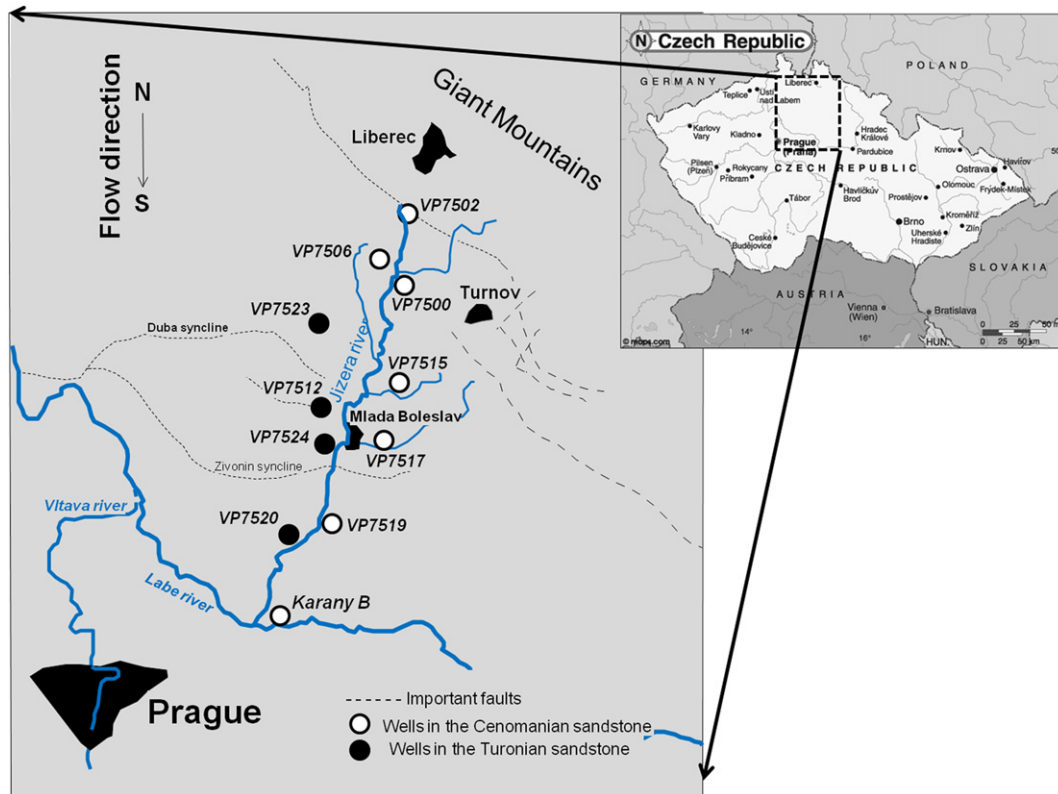


Fig. 1. Map of the Czech Republic, with the location of the studied area (square) and the wells.

In this study, we present a low-resolution climate record reconstructed from dissolved noble gases in groundwater from the Bohemian Cretaceous Basin, in the Czech Republic (Fig. 1). This record is combined with stable isotope (^{18}O , ^2H) measurements from groundwater, in order to gain information on the relationship between local and regional climate conditions and the groundwater recharge conditions. The Bohemian Basin lies in the corridor between the ancient Scandinavian ice sheet and the Alpine ice field and is hence a key region for understanding late Pleistocene climate and glacial development. Our results, in combination with other climate proxies, provide insight into the relation between climate conditions and groundwater recharge in Central Europe during the LGM and most of the Holocene.

2. Site description

The Bohemian Cretaceous Basin is the largest continuous sedimentary basin of the platform cover of the Bohemian Massif in central Europe (Fig. 1). The sediments of the basin accumulated from the early Cenomanian or even in the late Albian to the Santonian stage (Paces et al., 2009). The sediment thickness varies mostly between 200 and 400 m, but in some regions it reaches up to 1100 m (Paces et al., 2009). The sandstone layers are made of up to 99% of quartz sands. The most important aquifers are the regional shallow aquifer with a free and partly confined water table in Middle Turonian sandstones (Turonian aquifer - TA), and the regional deeper artesian aquifer in Cenomanian sandstones (Cenomanian aquifer - CA). These aquifers are separated by low-permeability marlstone and claystone aquitards (Paces et al., 2009). The base of the CA is formed by impermeable Permian rocks, schists and granitic rocks (aquifer scheme, Fig. 2) (Paces et al., 2009). Both aquifers have been intensively investigated over the past years and detailed characterizations in terms of groundwater mean residence

times, mixing processes and hydrogeochemical evolutions are well known (Paces et al., 2009; Purtschert et al., 2009).

The recharge area of both aquifers is located at the Jizera River and Labe River Basins, in the Northern part of the Czech Republic (Paces et al., 2009). For the CA, the recharge is constrained within a narrow elongated area at foothills of the Jizerské and Krkonoše (Giant) Mountains at the Czech Republic/Polish borders. During the LGM, numerous small glaciers developed in the inner valleys of the Giants Mountains (Ruzicka, 2004; Hauzenberger et al., 2009; Engel et al., 2010). These inner valleys are also located within the Jizera and Labe River Basins. The region is dominated by temperate climate conditions, with a mean annual precipitation rate of 1000 mm and a long term mean annual air temperature (MAAT) of 7.2 ± 2.0 °C (mean value of the period 1961–1996 at Liberec, Czech Meteorological Czech Hydrometeorological Institute, 2009).

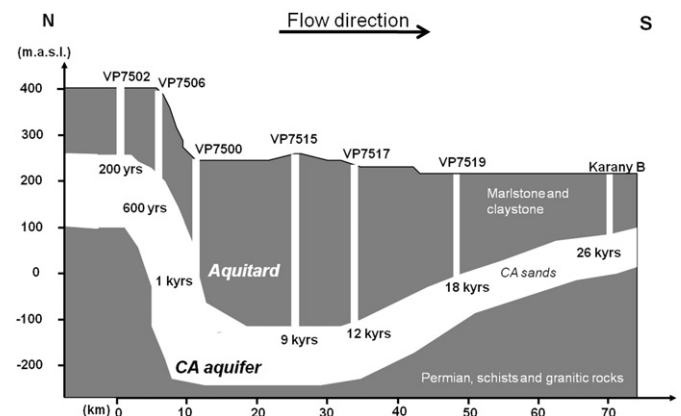


Fig. 2. Cross section of the CA aquifer, with location of the wells and the mean residence time of groundwater. The flow direction is also indicated.

Table 1

Altitude of the wells. The stable isotope ratios (‰ SMOW) in groundwater, the correspondent deuterium excess (d), and the mean groundwater ages are also shown.

Wells	Altitude (masl)	$\delta^{18}\text{O}$	$\delta^2\text{H}$	d	Groundwater age (yrs)	Recharge Period
Turonian sands aquifer (TA)						
VP7523	302.7	−10.0	−71.1	9.2	60 ± 20	Late Holocene (modern)
VP7512	305.6	−10.1	−72.6	8.1	200 ± 30	Pre-industrial Holocene
VP7524 ^a	221.3	−10.1	−73.1	7.3	4 ± 2	Late Holocene (modern)
VP7520	252.1	−9.5	−69.3	7.0	18 ± 4	Late Holocene (modern)
Cenomanian sands aquifer (CA)						
VP7502	385.2	−10.0	−69.6	10.4	200 ± 20	Pre-industrial Holocene
VP7506	369.5	−9.8	−66.7	11.4	600 ± 60	Pre-industrial Holocene
VP7500	345.4	−10.0	−68.8	11.3	970 ± 100	Pre-industrial Holocene
VP7515	228.8	−10.3	−71.2	10.8	7100 ± 700	Pre-industrial Holocene
VP7517	213.8	−10.7	−75.0	10.6	12000 ± 1200	Transition LGM/Holocene
VP7519	198.3	−11.5	−81.45	10.4	18000 ± 2000	LGP
Karany B ^b	198.0	−10.0	−72.1	8.0	26000 ± 3000	Late Pleistocene
Uncertainty		±0.5	±1.5	±1.6		

^a The sample is a mixture of two water components. The data correspond to the young water component (see [supplementary material](#)).^b The sample is a mixture of two water components. The data correspond to the old water component (see [supplementary material](#)).

3. Methods

Groundwater samples from four TA wells and seven CA wells (Table 1) were analyzed for hydrochemistry, stable isotope ratios ($\delta^{18}\text{O}$, $\delta^2\text{H}$), ^3H , carbon isotopes ($\delta^{13}\text{C}$, ^{14}C) and dissolved stable (He, Ne, Ar, Kr, Xe) and radioactive (^{37}Ar , ^{39}Ar , ^{85}Kr) noble gases. The late Holocene samples have been dated with the young and intermediate dating tracers ^3H , ^3He , ^{85}Kr and ^{39}Ar in former studies (Purtschert et al., 2009). The chronology of the early Holocene and the late Pleistocene samples was determined by radiocarbon dating. ^{14}C ages were obtained by geochemical modeling of carbon isotopes (Purtschert et al., 2009). The excess ^4He concentrations increase almost linearly with the ^{14}C ages and further validate the ^{14}C model ages for groundwater (See [supplementary material](#)). The NGTs and the excess air component (expressed as ΔNe) were calculated using published conceptual models and algorithms for air/water partition in groundwater (Aeschbach-Hertig et al., 1999, 2000; Klump et al., 2008b). In the case of one sample (Karany B, Table 1), for which bi-modal mixing was identified, the tracer data and age refer to the old groundwater fraction (Purtschert et al., 2009). For more details see [supplementary material](#).

4. Results and discussions

4.1. Groundwater age

Our investigated samples were either from the shallow TA that recharged in the late Holocene, or from the deeper and confined CA that recharged between the Late Pleistocene and pre-industrial Holocene (Table 1). The CA groundwater ages range between a few hundreds of years and 30000 years, and increase along the

flow direction North-South (aquifer scheme, Fig. 2) (Purtschert et al., 2009). One sample has a ^{14}C age of 18000 ± 2000 years, which may indicate recharge during times when ice covers developed in this region (Hauzenberger et al., 2009; Engel et al., 2010). However, this age has to be interpreted as a mean residence time and could be the result of water components that recharged before, during and/or after the LGM. The radiogenic ^4He concentrations showed an almost linear correlation with the ^{14}C model ages (See [supplementary material](#)) all along the investigated timescale (Late Pleistocene to Holocene).

4.2. Noble gas temperatures

The Holocene samples show mean NGTs (see [supplementary material](#)) between 5.1 and 8.9 °C (Table 2), which are in agreement with the present MAAT in the recharge zone (7.2 ± 1.0 °C) (Fig. 3a). Late Holocene or modern samples (water age < 100 yrs) have NGTs higher by 1.9 °C than pre-industrial Holocene samples (Fig. 3a). This temperature increase may be a consequence of rising air temperatures over the last 200–300 years, as it has been observed in other European aquifers (Corcho Alvarado et al., 2009).

The sample that recharged during the LGM (VP7519) exhibits a considerably lower NGT of just above the freezing point (0.8 ± 0.3 °C, Table 2 and Fig. 3a). Recharge under colder climatic conditions is also confirmed by depleted $\delta^{18}\text{O}$ and $\delta^2\text{H}$ values compared to the Holocene samples (Table 1, Fig. 4a). However, as groundwater recharge to the CA may have been interrupted by permafrost and ice covers during the LGM, this low NGT may not correspond to the lowest temperature at the LGM (Beyerle et al., 1998; Ma et al., 2004; Klump et al., 2008a; Blaser et al., 2010;

Table 2Noble gas concentrations in groundwater (in cm^3 per g of water). The results of the interpretation of the noble gases with the constant equilibration model are also shown.

Well	He (10^{-8})	$^3\text{He}/^4\text{He}$ (10^{-7})	Ne (10^{-7})	Ar (10^{-4})	Kr (10^{-8})	Xe (10^{-8})	χ^2	P (%)	NGT (°C)	Excess air (ΔNe)
Turonian sands aquifer (TA)										
VP7524	5.14 ± 0.03	14.4 ± 0.1	2.20 ± 0.02	4.02 ± 0.02	9.5 ± 0.1	1.37 ± 0.02	0.3	85.9	8.94 ± 0.08	7
VP7520	5.54 ± 0.07	17.1 ± 0.1	2.17 ± 0.02	4.15 ± 0.03	10.3 ± 0.1	1.50 ± 0.03	6.7	3.5	6.56 ± 0.43	2.7
Cenomanian sands aquifer (CA)										
VP7502	16.9 ± 0.2	7.8 ± 0.1	3.06 ± 0.02	4.87 ± 0.03	11.1 ± 0.1	1.55 ± 0.03	7.6	2.3	5.19 ± 0.49	43.5
VP7500	132 ± 2	9.6 ± 0.1	2.99 ± 0.02	4.74 ± 0.04	11.0 ± 0.1	1.59 ± 0.03	0.1	94.4	5.44 ± 0.06	40.3
VP7515	335 ± 4	10.6 ± 0.1	3.10 ± 0.02	4.68 ± 0.04	10.7 ± 0.1	1.61 ± 0.03	4.8	9.0	6.29 ± 0.39	46.4
VP7519	934 ± 12	75.6 ± 0.1	4.89 ± 0.04	6.25 ± 0.05	13.8 ± 0.2	1.94 ± 0.03	3.3	19.6	0.79 ± 0.33	117.5
Karany B ^a	1580 ± 16	7.7 ± 0.1	2.91 ± 0.03	4.64 ± 0.05	10.6 ± 0.1	1.50 ± 0.02	4.8	9.2	4.5 ± 0.35	70

^a This sample is a mixture of two water components. The data correspond to the old water component.

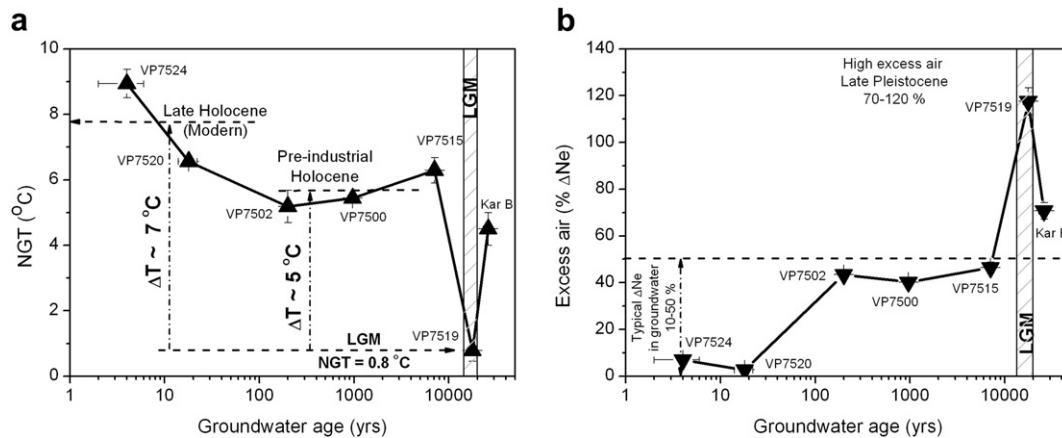


Fig. 3. a) Noble gas temperatures (NGT) and b) excess air values (expressed as ΔNe %) versus the age of groundwater (in logarithmic scale). The LGM is indicated. A mean atmospheric cooling of at least 5–7 °C is derived for the NGT data.

Varsanyi et al., 2011). Our reconstruction may have not fully captured the entire cooling during the LGM. The sample from Karany B infiltrated before the LGM, under warmer climate conditions than during the LGM; but still colder than in the Holocene (Table 2, Fig. 3a).

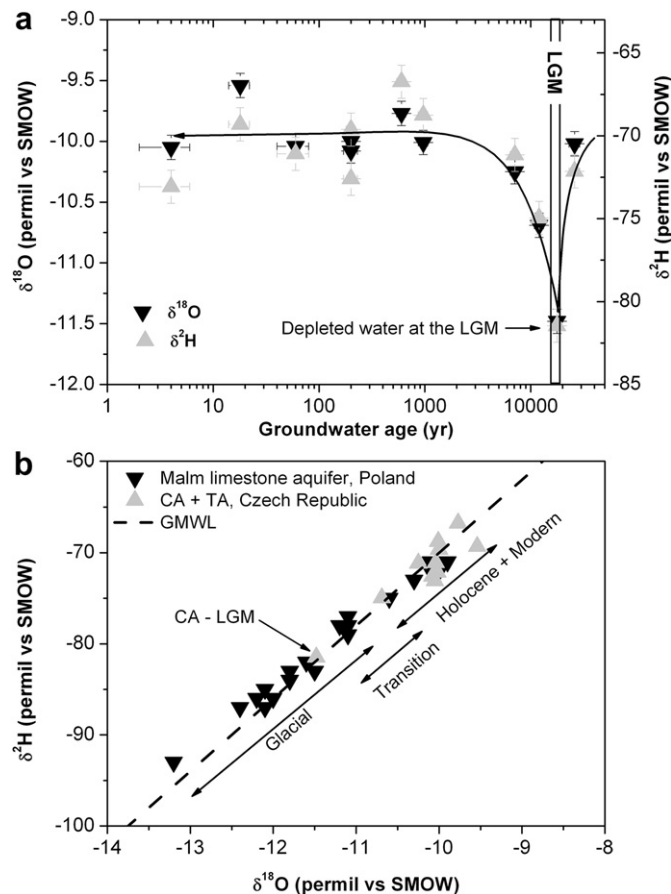


Fig. 4. a) $\delta^{18}\text{O}$ and $\delta^2\text{H}$ (‰ SMOW) versus the ^{14}C age of groundwater from the Czech aquifers. The LGM is indicated. A shift to more depleted ^{18}O and ^2H is observed for the glacial sample, which reflects a colder atmospheric temperature. b) $\delta^{18}\text{O}$ versus $\delta^2\text{H}$ (‰ SMOW) for the Czech aquifers and for the Malm Limestone aquifer (MA, Poland; Zuber et al., 2004). GMWL – global meteoric water line.

The NGT record reveals a temperature rise of at least $5.1 \pm 0.6^\circ\text{C}$ from the LGM to the pre-industrial Holocene (CA samples), and a temperature raise of at least $7.0 \pm 1.2^\circ\text{C}$ from the LGM to the late Holocene (TA samples). This glacial/interglacial warming is comparable to the mean temperature increase of 4.5–7 °C found in other European aquifers (Rudolph et al., 1984; Beyerle et al., 1998; Zuber et al., 2000, 2004; Kipfer et al., 2002) and argues for an homogeneous continental cooling (Fig. 5). For example, LGM temperatures 5–7 °C lower than for the Holocene were estimated in various aquifer systems of Germany based on the noble gases dissolved in groundwater (Rudolph et al., 1984). Similarly, a glacial/interglacial warming of at least 5 °C was reported in the Glattal Valley in Switzerland (Beyerle et al., 1998). Temperature shifts of 4.5–5 °C have been found in two aquifer systems in Poland (Zuber et al., 2000, 2004).

Larger LGM – Holocene temperature shifts between 8.6 and 9.5 °C have been reported for the Great Hungarian Plain (Stute and Deák, 1989) and the Ledo-Paniselian aquifer in Belgium (Blaser et al., 2010). During the LGM, recharge into these aquifers was affected or even interrupted by permafrost and ice cover (Stute and Deák, 1989; Blaser et al., 2010). Therefore, the noble gas method may have not captured the whole cooling at the LGM. Due to the higher modern MAAT in the recharge areas of these two sites compared to our studied site, the recharge temperatures obtained from the noble gases would result in a larger LGM – Holocene shift compared to our site.

As already pointed out by other authors, the preservation of a climatic signal in a groundwater system is strongly related to the specific hydrodynamic characteristic of the system (Stute and Schlosser, 1993; Castro et al., 2007). Furthermore, the smoothing of the climate signal by dispersion would also be different in each groundwater system. This phenomenon may play an important role for explaining the differences in the LGM/Holocene temperature shifts reconstructed from the above aquifer systems (Fig. 5). However, due to the lack of information about hydrodynamic characteristic of the systems, further conclusions cannot be made.

Our reconstructed glacial/interglacial temperature shift is lower than the values found in other reconstructions from different low resolution proxy records of the Czech Republic. Several reconstructions have consistently estimated a glacial/interglacial warming of 7–11 °C in the Czech Republic (Bodri and Cermak, 1997; Safanda et al., 2000; Safanda and Rajver, 2001). Analysis of ice accumulation in the Giant Mountains concluded that the mean annual

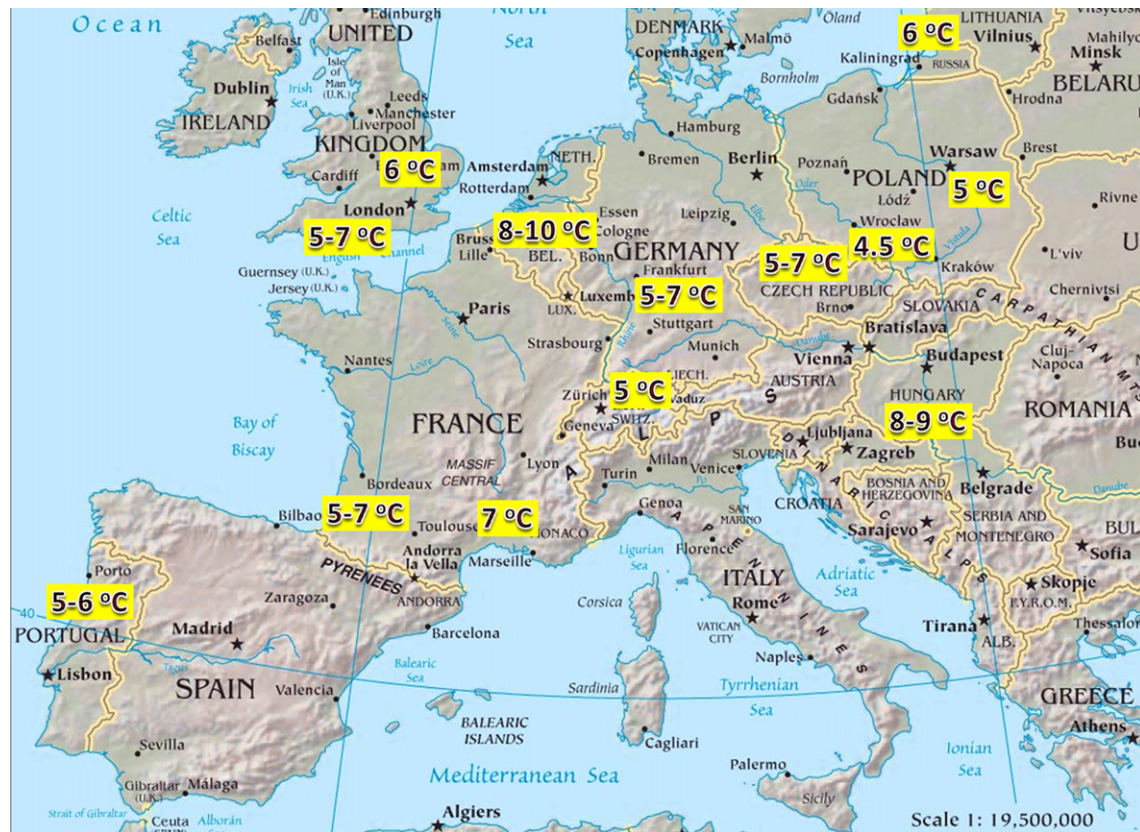


Fig. 5. Glacial/interglacial temperature warming at different sites in Europe reconstructed from groundwater data (Andrews and Lee, 1979; Bath et al., 1979; Rudolph et al., 1984; Stute and Deák, 1989; Beyerle et al., 1998; Elliot et al., 1999; Zuber et al., 2000; Condeso de Melo et al., 2001; Huneau et al., 2001; Zuber et al., 2004; Blaser et al., 2010).

temperature decreased maximum 8–9 °C during the LGM (Hauzenberger et al., 2009).

4.3. Excess air component – ΔNe

High amounts of excess air was found in the samples that recharged before and during the LGM ($\Delta Ne = 70$ –120%, Table 2, Fig. 3b). ΔNe values typical for recent groundwater range between 10 and 50% (Wilson and McNeill, 1997; Kipfer et al., 2002; Klump et al., 2008b). Two processes are identified to generate high excess air loads in porous media: (i) increased groundwater table fluctuations and increased hydraulic loading i.e. due to more frequent intense rain events (Wilson and McNeill, 1997; Klump et al., 2007; Loose et al., 2009) or the development of ice covers (Klump et al., 2008a), and (ii) admixture of water from ice melting under a glacier (sub-glacial meltwater) which is known to contain large amounts of excess air ($\Delta Ne > 500\%$) as a result of trapped air bubbles in glacial ice (Aeschbach-Hertig et al., 1999; Vaikmaa et al., 2001; Loose et al., 2009). It is important then to distinguish between these two processes because a small amount of meltwater would induce excess air which can be erroneously interpreted as a significant increase of recharge rates (Klump et al., 2007).

During the LGM, the climate was dry and the air temperatures were near the freezing point (NGT: 0.8 ± 0.3 °C, Table 2). A large number of small glaciers developed in the inner valleys of the Giant Mountains (Hauzenberger et al., 2009; Engel et al., 2010); and parts of the Czech Republic were covered by discontinuous permafrost (Czudek, 1992). Therefore, a large increase in the frequency of recharge events (process(i)) due to a drastic increase in precipitation was very unlikely.

To our knowledge, plausible mechanisms facilitating meltwater infiltration containing high excess air from the base of a glacier haven't been experimentally identified. However, the high hydrostatic pressures prevailing at the base of a glacier (Hoaglund et al., 2004; Person et al., 2007) holds the excess air in solution. The excess air in the glacial sample ($\Delta Ne: 120\%$) is several times lower than the typical values found in pure glacial meltwater ($\Delta Ne > 500\%$, Vaikmaa et al., 2001), which argues against recharge of pure meltwater. Moreover, the glacial sample does not show highly depleted stable isotope ratios characteristic of pure meltwater. This is an additional argument against the recharge of pure meltwater (Vaikmaa et al., 2001; Klump et al., 2008a).

Infiltration of some sub-glacial meltwater (process ii) cannot however be completely excluded. If meltwater contains a ΔNe of minimum 500% (Vaikmaa et al., 2001), then it can be deduced that maximum 17% of meltwater infiltrated into the aquifer (due to mixing with water that contains a ΔNe of ca. 40%, Table 1). A significant amount of another water component (>83% of the total) would be also recharging the aquifer. Therefore, the input of sub-glacial meltwater does at the most partly explain the elevated ΔNe values.

In numerous sites worldwide, the developments of ice covers and permafrost have induced significant changes in the hydraulic conditions during groundwater recharge (Bath et al., 1979; Bense and Person, 2008; Klump et al., 2008a; Blaser et al., 2010; McIntosh et al., 2010). In some aquifers, the ice covers have even prevented groundwater infiltration (Bath et al., 1979; Beyerle et al., 1998; Blaser et al., 2010). The high ΔNe observed in the glacial sample may then indicate an abrupt change in the recharge dynamics of groundwater from the end of the Pleistocene to the Holocene due to the rapid progression and retreat of the ice covers

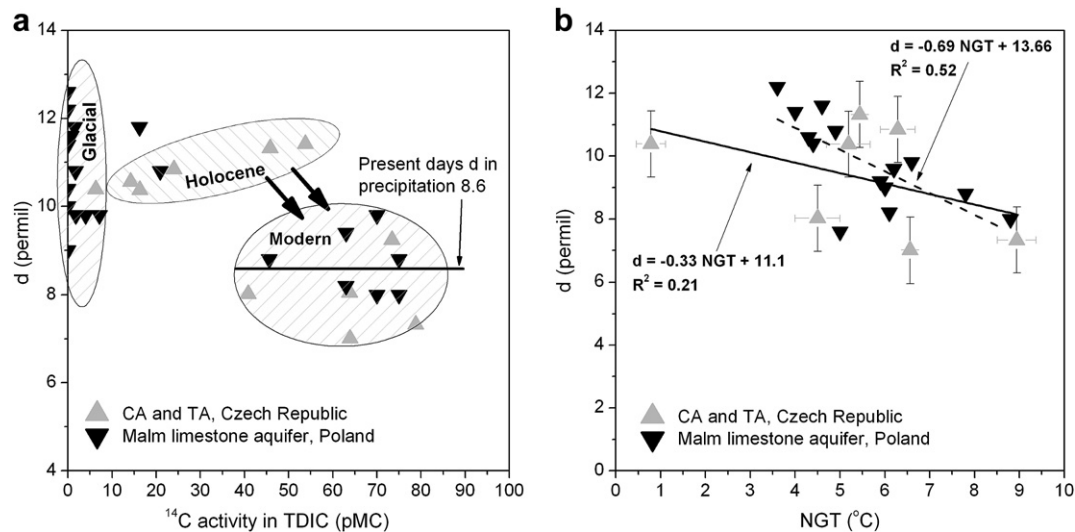


Fig. 6. Variation of $-d$ (‰) as a function of: a) the ^{14}C activity (expressed as percent modern carbon – pMC) in the Czech aquifers CA and TA, and the Malm Limestone aquifer (MA) in Poland (Zuber et al., 2004); and b) the NGT from the CA and TA.

and permafrost in the recharge area of the aquifer. Changes in the recharge dynamics of groundwater due to the advance of the mountain glaciers, and potentially also some short periods of ice retreat before the LGM, may possibly be the reason of the relatively high ΔNe (70%) observed in the sample that infiltrated before the last glacial period (Karany B, Table 2).

4.4. Stable isotopes

The stable isotope ratios $\delta^2\text{H}$ and $\delta^{18}\text{O}$ of the Turonian and Cenomanian samples lie close to the global meteoric water line (Fig. 4b; GMWL: $\delta^2\text{H} = 8.1 \delta^{18}\text{O} + 10.8$; Rozanski et al., 1992), indicating that groundwater is of meteoric origin and that isotope exchange within the aquifer is of minor importance. Slightly depleted isotope ratios of samples from the LGM ($\delta^2\text{H}$ by 12.4‰ ; $\delta^{18}\text{O}$ by 1.5‰) and the glacial/interglacial transition ($\delta^2\text{H}$ by 5.9‰ ; $\delta^{18}\text{O}$ by 0.7‰) reflect the lower air temperatures that prevailed during these periods. The stable isotope ratios match well the continental gradients observed in recent infiltrated waters ($\delta^2\text{H} \sim -70\text{‰}$; $\delta^{18}\text{O} \sim -10\text{‰}$) and palaeowaters ($\delta^2\text{H} \sim -80\text{‰}$; $\delta^{18}\text{O} \sim -11.5\text{‰}$) in Europe (Rozanski, 1985). Furthermore, the $\delta^2\text{H}$ and $\delta^{18}\text{O}$ values are indistinguishable from the values measured in groundwater from the Malm limestone aquifer, south Poland (Fig. 4b). These results evidence that groundwater from the investigated aquifers may have preserved at least regional $\delta^2\text{H}$ and $\delta^{18}\text{O}$ isotope signatures.

The parameter deuterium excess d (defined as $d = \delta^2\text{H} - 8 \cdot \delta^{18}\text{O}$; Dansgaard, 1964) shows some distinctive change in response to the climatic shift from the LGM to the Holocene. Whereas d does stay nearly constant from the LGM to the pre-industrial Holocene, d decreases significantly from the pre-industrial Holocene to the modern Holocene samples (Fig. 6a). A similar pattern was also recorded in the Malm Limestone aquifer in south Poland (Fig. 6a), suggesting at least a shift in deuterium excess over the last few centuries. Deuterium excess in precipitation decreases in response to enhanced humidity over the oceanic source, as a result of reduced moisture recycling or stronger sub-cloud evaporation at the place of recharge (Jouzel et al., 1997; Froehlich et al., 2002). The relative strength of these processes determines the d that is finally observed in groundwater. In our study site, d shows a negative correlation with the NGTs, as the one observed in the Malm limestone aquifer (Fig. 6b). Although weak, the correlations follow the same seasonal variation of d observed in modern precipitation

(Jouzel et al., 1997; Froehlich et al., 2002). Today, for the warmer months, which correspond to the periods of higher humidity, minimums of d are observed in precipitation and vice versa. The temporal decrease of d from the pre-industrial Holocene to present days (Fig. 6a) may be linked to the increase of the air temperatures and probably also to an increase of the humidity at the source region of precipitation.

5. Conclusions

The studied site is a key region for understanding late Pleistocene climate and glacial development in a periglacial region beyond the limits of the Scandinavian and the alpine ice sheets. ^{14}C ages, $\delta^2\text{H}$, $\delta^{18}\text{O}$ and NGT indicate consistently that groundwater recharged during the last glacial period. A temperature increase in groundwater from the LGM to the Holocene of at least $5\text{--}7\text{ }^{\circ}\text{C}$ was obtained, which is similar to the warming reported in other studies in Europe. Our study provided further evidences of the large impact of ice development during the LGM on groundwater recharge conditions.

Author contributions

José A. Corcho Alvarado and Roland Purtschert were in charge of the field work (sampling of gas, soil, dissolved carbonate and groundwater), and the preparation and measurement of the radioactive tracers (^{85}Kr , ^{39}Ar , ^3H , ^{14}C). Markus Leuenberger was in charge of the analysis of the stable isotopes of the water ^{18}O and ^2H . Rolf Kipfer was in charge of the analysis of the stable noble gases. Tomas Paces was a field hydrogeologist and interpreted the hydrogeological situation of the investigated area. All the authors contributed similarly to the preparation of the manuscript.

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Appendix. Supplementary material

Supplementary data associated with this article can be found, in the online version, at doi:10.1016/j.quascirev.2011.09.003.

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